

LysoStain™ Acidic Organelle Stain

Catalog #	Description
135-1330	LysoStain Blue 353/442 Acidic Organelle Stain
135-1331	LysoStain Green 450/505 Acidic Organelle Stain
135-1332	LysoStain Yellow 542/556 Acidic Organelle Stain
135-1333	LysoStain Orange 575/597 Acidic Organelle Stain
135-1334	LysoStain Red 596/619 Acidic Organelle Stain
135-1335	LysoStain Deep Red 636/650 Acidic Organelle Stain

For research purposes only.

Description

LysoStain is an easy-to-use probe-based solution for visualizing lysosomes and other acidic organelles in mammalian cells by fluorescence microscopy.

Lysosomes are spherical acidic organelles containing hydrolytic enzymes that digest many different biomolecules and cellular debris. The acidic environment (pH 4.5–4.8) in these organelles is essential to their function and is preserved via the activity of proton pumps and chloride ion channels. Lysosomes help maintain cellular homeostasis and play a fundamental role in preserving cell health.

Bio-Rad's LysoStain takes advantage of a fluorogenic probe that permeates the plasma membranes of live cells and selectively accumulates in acidic organelles (Figure 1). The probe's fluorescence is significantly enhanced on entering acidic compartments.

The LysoStain family of products can be used with both suspension and adherent cells, as well as proliferating and nonproliferating cells.

LysoStain Acidic Organelle Stains are available in six different excitation/emission wavelength combinations (Table 1) to enable multicolor experiments using the most common fluorescence cell imagers and microscopes.

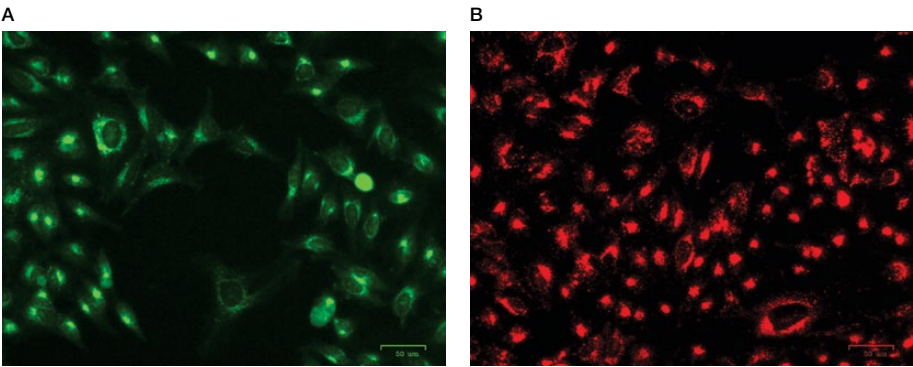


Fig. 1. Staining of acidic organelles with 1x LysoStain Green 450/505 Detection Kit (A) and 0.05x LysoStain Orange 575/597 Detection Kit (B). HeLa cells were stained with LysoStain and visualized with the ZOE™ Fluorescent Cell Imager.

Contents and Storage

Each box contains one vial of LysoStain (100 µl of 500x stain per vial). Store LysoStain at -20°C. Protect from light and avoid repeated freeze-thaw cycles.

Ordering Information

Use Table 1 to select the appropriate LysoStain for any given cell imaging experiment.

Table 1. Ordering information.

Catalog Number	Description	Optimal Channel on ZOE Fluorescent Cell Imager	Excitation, nm	Emission, nm
135-1330	LysoStain Blue 353/442	Blue	353	442
135-1331	LysoStain Green 450/505	Green	450	505
135-1332	LysoStain Yellow 542/556	Red	542	556
135-1333	LysoStain Orange 575/597	Red	575	597
135-1334	LysoStain Red 596/619	Red	596	619
135-1335	LysoStain Deep Red 636/650	—	636	650

Protocol

This protocol provides guidelines and should be modified according to your specific needs.

Preparing the Working Solution

1. Warm LysoStain to room temperature and centrifuge briefly before opening.
2. Prepare a working solution of 1x LysoStain by diluting the stock solution 1:500 in cell growth medium.

Note 1: 20 µl of LysoStain makes 10 ml of 1x working solution, which is sufficient for one 96-well plate. Aliquot and store unused stock solution below -20°C .

Note 2: Optimal concentration of LysoStain varies depending on the specific application. Staining conditions may be modified according to the particular cell type and the permeability of the cells or tissues to the probe.

Preparing and Staining Adherent Cells

1. Grow the cells according to your experimental procedure. When cells reach the desired confluence, replace the cell growth medium with an equal volume of 1x LysoStain (see Preparing the Working Solution section).
2. Incubate the cells in a 37°C , 5% CO_2 incubator for 30 min.
3. Wash the cells twice with prewarmed (37°C) phosphate buffered saline (PBS) or buffer of your choice. Keep the cells in growth medium.
4. Observe the cells using a ZOE Fluorescent Cell Imager or any fluorescence microscope equipped with an appropriate filter set and light source.

Preparing and Staining Suspension Cells

1. Spin down the cells and replace the cell growth medium with an equal volume of 1x LysoStain (see Preparing the Working Solution section). Resuspend the cells gently and thoroughly.
2. Incubate the cells in a 37°C , 5% CO_2 incubator for 30 min.
3. Wash the cells twice with prewarmed (37°C) PBS or buffer of your choice.
4. Keep the cells in growth medium; mount an aliquot on a slide under a coverslip if desired.
5. Observe the cells using a ZOE Fluorescent Cell Imager or any fluorescence microscope equipped with an appropriate filter set and light source.

Visit bio-rad.com/web/LysoStain for more information.

